



## TOPIC 2

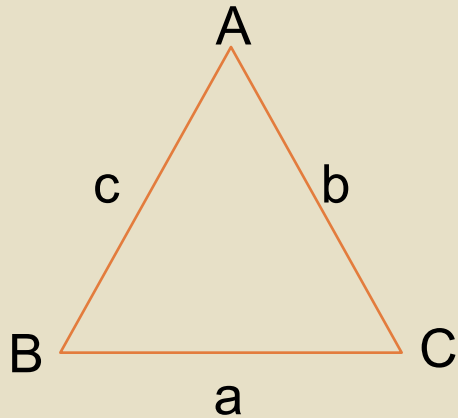
# TRIGONOMETRY

## **2.2 TRIANGLE PROBLEMS**

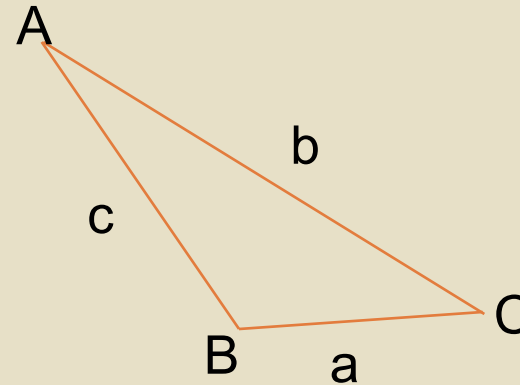
Triangle problems using:

1. Sine formula
2. Cosine formula
3. Pythagoras Theorem

# 1. SINE FORMULA



OR



Sine Formula:

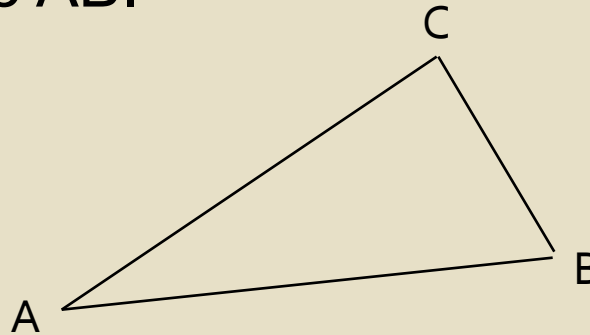
$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

OR

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

# EXAMPLE

$\angle A = 30^\circ$ ,  $\angle B = 85^\circ$  and  $BC = 6$  m. Compute distance  $AC$ ,  $\angle C$  and distance  $AB$ .



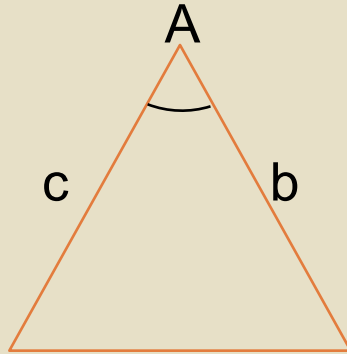
$$AC / \sin B = BC / \sin A$$

$$AC / \sin 85^\circ = 6 / \sin 30^\circ$$

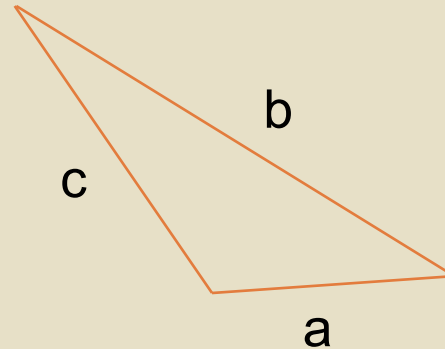
$$AC = [6 / \sin 30^\circ] \times \sin 85^\circ$$

$$\underline{\underline{AC = 11.954 \text{ m}}}$$

## 2. COSINE FORMULA



OR

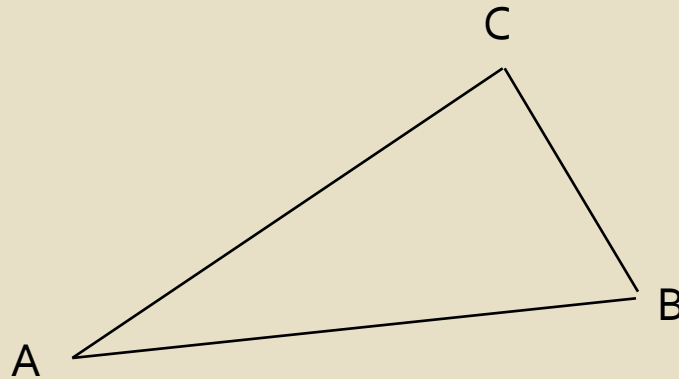


Cosine Formula:

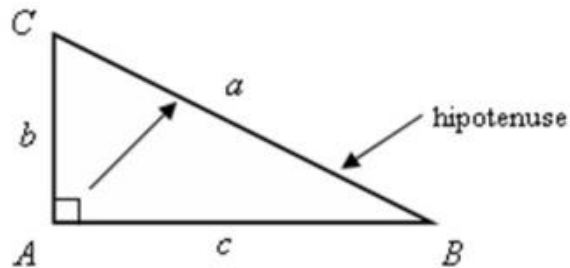
$$a^2 = b^2 + c^2 - 2bc \cos A$$

# EXAMPLE

$\triangle ABC$  as  $b = 5$  m,  $c = 6$  m,  $\angle A = 52^\circ$ . Calculate the distance of  $a$ .



### 3. PYTHAGORAS THEOREM



Right Angle Triangle

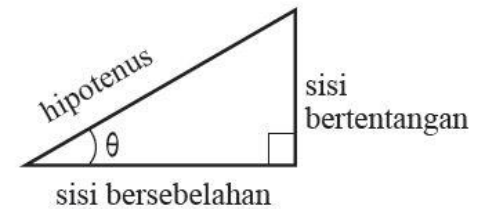
$$a^2 = b^2 + c^2$$

$$a = \sqrt{b^2 + c^2}$$

$$b = \sqrt{a^2 - c^2}$$

$$c = \sqrt{a^2 - b^2}$$

OR



Oleh itu,  $\sin \theta = \frac{\text{sisi bertentangan}}{\text{hipotenuse}}$

$$\cos \theta = \frac{\text{sisi bersebelahan}}{\text{hipotenuse}}$$

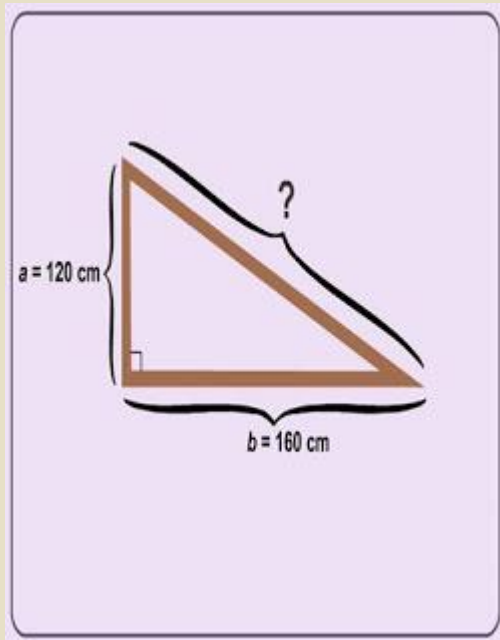
$$\tan \theta = \frac{\text{sisi bertentangan}}{\text{sisi bersebelahan}}$$

**$\sin \theta^\circ = \text{Opposite} \div \text{Hypotenuse}$**

**$\cos \theta^\circ = \text{Adjacent} \div \text{Hypotenuse}$**

**$\tan \theta^\circ = \text{Opposite} \div \text{Adjacent}$**

# EXAMPLE



$$c^2 = a^2 + b^2$$

$$120^2 + 160^2 = c^2$$

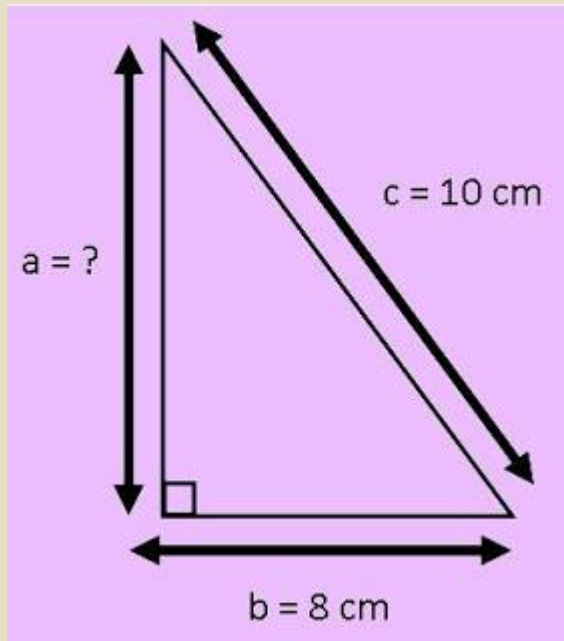
$$14400 + 25600 = c^2$$

$$c^2 = 40000$$

$$\sqrt{c^2} = \sqrt{40000}$$

$$c = 200 \text{ cm}$$

# Exercise 1



$$c^2 = a^2 + b^2$$

$$10^2 = a^2 + 8^2$$

$$a^2 = 10^2 - 8^2$$

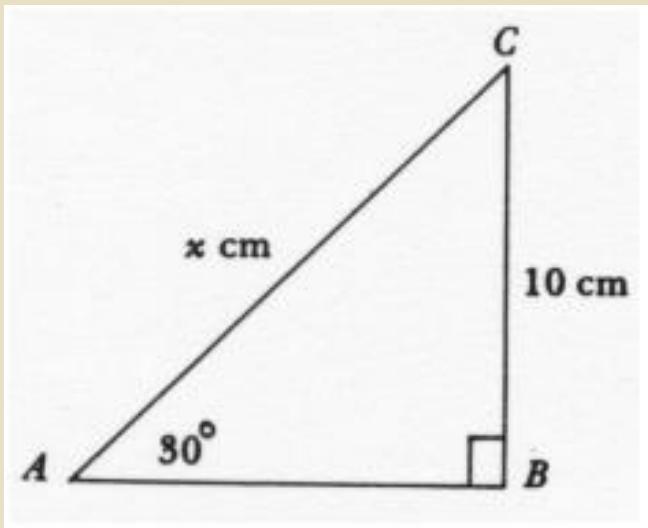
$$= \sqrt{100 - 64}$$

$$a = \sqrt{36}$$

$$\mathbf{a = 6.000\ m}$$

## Exercise 2

Compute distance of AC



$$\sin \theta^\circ = \text{Opposite} \div \text{Hypotenuse}$$

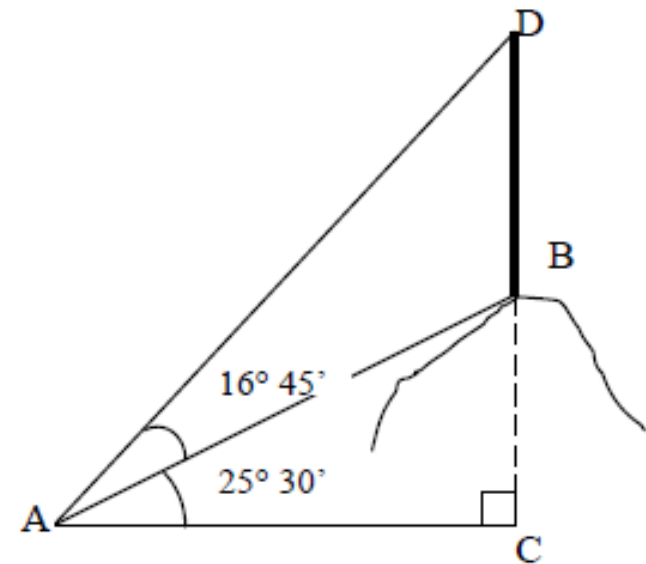
$$\sin 30^\circ = 10 / AC$$

$$AC = 10 / \sin 30^\circ$$

$$\underline{AC = 20 \text{ cm}}$$

### Contoh 3.2

Seorang pengukur telah membuat cerapan dari setesen A ke sebatang tiang yang terletak di kawasan tinggi. Bacaan sudut pugak ke kaki tiang(B) ialah  $25^\circ 30'$  dan sudut pugak B ke puncak tiang(D) ialah  $16^\circ 45'$ . Jarak AB ialah 80.0m (Rajah 3.3).

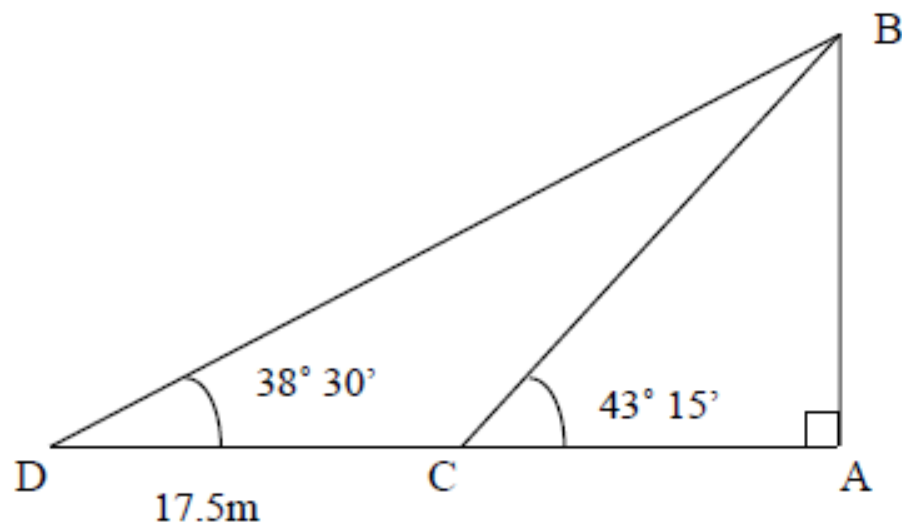


*Rajah 3.3*

Hitungkan :

- i. Jarak mendatar AC.
- ii. Tinggi tiang tersebut.

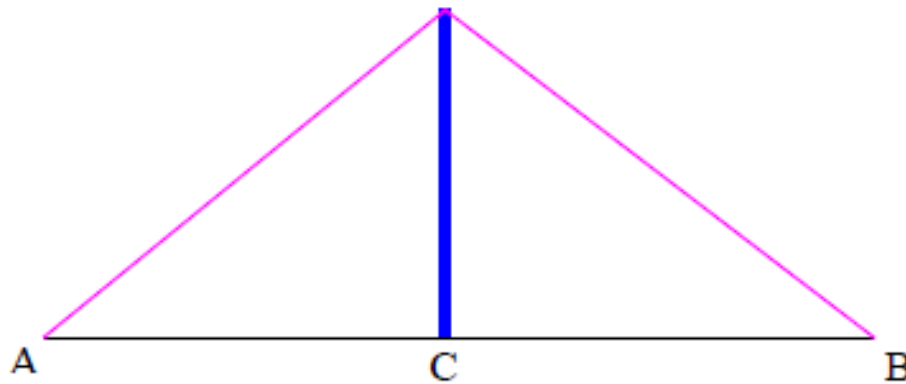
**Jawapan: 31.147 m**



*Rajah 3.5*

Merujuk kepada *rajah 3.5* dan data-data yang diberi, hitungkan ketinggian  $AB$  di mana jarak  $AD$  tidak dapat diukur secara terus.

**Jawapan: 90.141 m**



*Rajah 3.12*

Merujuk *rajah 3.12*, sebatang tiang titik C yang berada diantara titik cerapan A dan cerapan B. Sudut dongak dari A ke puncak tiang ialah  $14^{\circ} 02'$  dan dari titik B ialah  $18^{\circ} 26'$ . Jika tinggi tiang 30m, hitungkan jarak AB.

**Jawapan: 210.030m**